AS · Unit 2 · WCH12 Energetics, Group Chemistry, Halogenoalkanes and Alcohols

Intermolecular forces

Covalent bonds hold the atoms together inside a molecule. Much weaker forces act between neighbouring molecules, and those forces decide whether a substance is a gas, a liquid or a solid, how hot it must be to boil, and what it will dissolve in. This topic explains the three kinds of intermolecular force, how to tell which ones a substance has, and how to use them to explain real data.

Forces between molecules

When a simple molecular substance melts or boils, its molecules move apart, but the molecules themselves stay intact. Boiling water gives steam made of H_2O molecules, not hydrogen and oxygen. So the energy needed to boil a molecular substance is the energy needed to overcome the **intermolecular forces**, and the stronger those forces are, the higher the boiling temperature.

Force	Between which molecules?	Caused by	Relative strength
London forces (instantaneous dipole–induced dipole)	all molecules, and atoms of the noble gases	temporary, constantly changing dipoles	weak for small molecules; grow with the number of electrons
Permanent dipole–dipole forces	polar molecules	permanent δ+ and δ− charges lining up	usually smaller than the London forces in the same molecule
Hydrogen bonds	molecules with H bonded to N, O or F	a strongly δ+ hydrogen attracted to a lone pair on N, O or F	the strongest intermolecular force

KEY IDEA

All three are far weaker than covalent bonds. Even a hydrogen bond is only a small fraction of the strength of the O–H bond inside a water molecule. Forces add up: a molecule that forms hydrogen bonds also has London forces and permanent dipole–dipole forces.

London forces

The electrons in an atom or molecule are always moving. On average they are spread evenly, but at any instant there may be slightly more electron density on one side than the other. That side is momentarily δ−, the other side

δ+: an **instantaneous dipole**.

This dipole affects the next molecule. Its δ− side repels the electrons of the neighbour, and its δ+ side attracts them, so the neighbour's electron cloud shifts too: an **induced dipole**. The δ+ of one molecule now faces the δ− of the next, and they attract. A moment later the dipoles have changed, but the induced dipoles always follow the instantaneous one, so on average there is a net attraction. These attractions are **London forces**. They act between **all** molecules, including non-polar ones and single atoms of the noble gases.

Two things control the strength of London forces:

- **The number of electrons.** More electrons means a larger electron cloud that is held less tightly and distorts more easily, so the instantaneous and induced dipoles are larger. This is why boiling temperatures rise down group 0 and down group 7, and along any homologous series.
- **The area of contact between molecules.** Long, thin molecules can lie alongside each other and touch along their whole length. Compact, branched or nearly spherical molecules touch over a smaller area, so for the same number of electrons the London forces are weaker. (See the alkane isomers on the alkanes page.)

Halogen	Electrons per molecule	Boiling temperature	State at room temperature
F_2	18	−188 °C	gas
Cl_2	34	−34 °C	gas
Br_2	70	59 °C	liquid
I_2	106	184 °C	solid

Permanent dipole–dipole forces

When two atoms with different electronegativities share electrons, the bonding pair is pulled towards the more electronegative atom. The bond is **polar**, with a permanent δ− on one end and δ+ on the other. A molecule is **polar** if it has polar bonds whose dipoles do not cancel out.

- HCl, H_2S, propanone and halogenoalkanes such as chloromethane are polar molecules.
- CO_2 and CCl_4 have polar bonds, but their shapes are symmetrical (linear and tetrahedral), so the bond dipoles cancel. These molecules are **non-polar** and have London forces only.

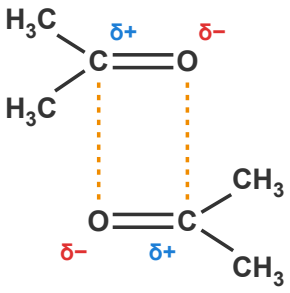
Polar molecules line up so that the δ+ end of one is near the δ− end of the next. The attraction between these permanent dipoles is a **permanent dipole–dipole force**. It acts in addition to the London forces.

hydrogen chloride

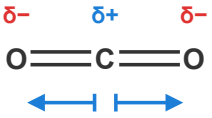


δ− of one molecule attracts δ+ of the next

propanone



carbon dioxide



bond dipoles cancel:
a non-polar molecule

Orange dotted lines: permanent dipole–dipole attractions between polar molecules. CO_2 has polar C=O bonds, but its linear shape makes the two dipoles cancel, so it has London forces only.

COMMON MISTAKE

Do not assume the dipole–dipole forces are the biggest part. For most polar molecules the London forces are larger. HCl, HBr and HI become **less** polar down the group, yet their boiling temperatures **rise**, because the number of electrons, and so the strength of the London forces, increases.

Hydrogen bonds

A **hydrogen bond** is a strong intermolecular attraction between a hydrogen atom bonded to nitrogen, oxygen or fluorine and a **lone pair** on a nitrogen, oxygen or fluorine atom in another molecule. Two features make it much stronger than an ordinary dipole–dipole force:

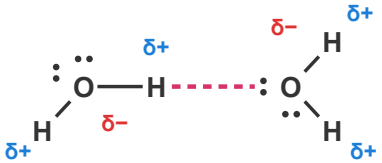
- N, O and F are the most electronegative elements, so the N–H, O–H or F–H bond is very polar and the hydrogen carries a large δ+ charge.
- A hydrogen atom has no inner electrons: pulling its one electron away leaves an almost bare proton, which is tiny, so the lone pair of another molecule can get very close to it.

hydrogen fluoride



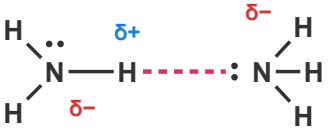
F–H···F: 180°

water



O–H···O: 180°

ammonia



N–H···N: 180°

hydrogen bond, from δ+ H to a lone pair

In each case the hydrogen bond (pink) runs from a δ+ hydrogen to a lone pair on N, O or F, in a straight line. Only one hydrogen bond is drawn in each pair; real liquids form a network of them.

EXAM TIP

Drawing hydrogen bonds for full marks

Show the δ+ on the H and the δ− on the N, O or F atoms; draw the **lone pair** on the atom accepting the hydrogen bond; draw the hydrogen bond as a dashed line from the H to that lone pair; and make the X–H···X arrangement a straight line, **180°**. It is linear because there are two pairs of electrons around the hydrogen (the bonding pair and the lone pair it is attracted to), and they repel each other to be as far apart as possible.

To **predict** whether a substance forms hydrogen bonds, look for an H atom bonded directly to N, O or F. Water, ammonia, hydrogen fluoride, alcohols (–OH), carboxylic acids (–COOH) and amines ($–NH_2$) all qualify. Fluoromethane, CH_3F, does not: its hydrogens are bonded to carbon. Neither do propanone, ethers or aldehydes, although their oxygen lone pairs can **accept** hydrogen bonds from water, which helps them dissolve.

The anomalous properties of water

Unusually high boiling temperature

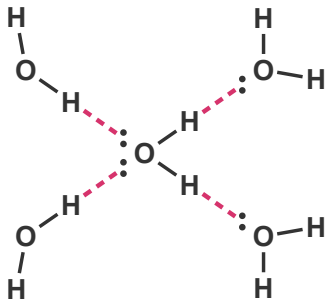
Down each group of the periodic table the hydrides get more electrons, so London forces increase and boiling temperatures rise. Water, ammonia and hydrogen fluoride break this pattern: they have the fewest electrons in their groups, yet boil far higher than the next hydrides down, because they form hydrogen bonds. Methane, in group 14, has no hydrogen bonds and fits the trend.

Why does water boil higher than both HF and NH_3? Count the hydrogen bonds each molecule can make. A water molecule has **two** δ+ hydrogens and **two** lone pairs on oxygen, so on average it forms two hydrogen bonds (and takes part in up to four). HF has three lone pairs but only one hydrogen, and NH_3 has three hydrogens but only one lone pair, so each forms on average only one hydrogen bond per molecule. The hydrogen bonds in HF are individually stronger than those in NH_3, because F is more electronegative than N, so HF boils higher than NH_3.

Ice is less dense than water

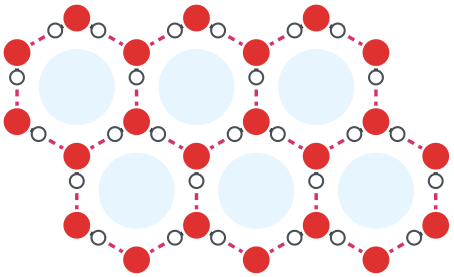
Nearly every substance is denser as a solid than as a liquid. Water is the exception, which is why ice floats.

one molecule, four hydrogen bonds



in three dimensions the four bonds point to the corners of a tetrahedron

oxygen atoms in ice: an open lattice



red: O · white: H · pink dashes: hydrogen bonds
the pale circles mark the empty space in each ring

Left: in ice each molecule donates two hydrogen bonds and accepts two. Right: the hydrogen bonds lock the oxygen atoms into hexagonal rings with a lot of empty space (each oxygen also bonds to a molecule above or below the page). Melting breaks some hydrogen bonds and the molecules move closer together.

Why ice floats

1. In ice, every water molecule is hydrogen-bonded to four others, arranged tetrahedrally around it: two bonds through its own hydrogens and two through the lone pairs on its oxygen.
2. The hydrogen bonds have a fixed length and direction, so they hold the molecules in a rigid, **open lattice** with a lot of empty space, and the molecules further apart than in liquid water.
3. When ice melts, some hydrogen bonds break, the open lattice collapses, and the molecules can pack closer together. So the same mass takes up less volume: liquid water is denser than ice.

Ice floating has a big consequence: lakes freeze from the top down, and the ice layer insulates the water below, so aquatic life survives the winter.

Explaining trends in boiling temperatures

Whenever you explain a boiling temperature, decide which forces are present, decide which one makes the difference, and say that more energy is needed to overcome stronger forces **between the molecules**.

Alkanes: chain length and branching

Alkanes are non-polar, so they have London forces only. Each extra CH_2 adds 8 electrons, so boiling temperatures rise steadily with chain length. Branched isomers have the same number of electrons but a smaller area of contact,

so they boil lower: pentane boils at 36 °C, 2,2-dimethylpropane at 10 °C. The alkanes page has an interactive plot of both effects.

Alcohols are much less volatile than alkanes

A **fair** comparison uses molecules with the same number of electrons, so that the London forces are about the same. Then any difference comes from the other forces. Ethanol and propane both have 26 electrons, but ethanol boils at 78 °C and propane at −42 °C. Ethanol molecules form hydrogen bonds through their –OH groups, and a lot more energy is needed to overcome them. "Low volatility" just means the liquid does not evaporate easily, which goes with a high boiling temperature.

The hydrogen halides

	HF	HCl	HBr	HI
Electrons per molecule	10	18	36	54
Boiling temperature	20 °C (293 K)	−85 °C (188 K)	−67 °C (206 K)	−35 °C (238 K)

Explaining the boiling temperatures of HF to HI

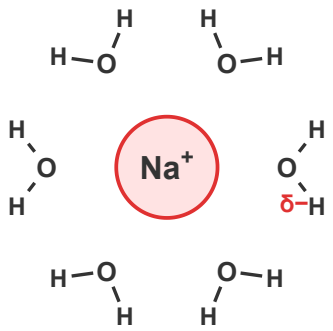
1. **HCl → HBr → HI:** the number of electrons increases (18, 36, 54), so the London forces between the molecules get stronger.
2. The molecules actually become **less** polar down the group, because the difference in electronegativity between H and the halogen gets smaller. So the permanent dipole–dipole forces get weaker.
3. The boiling temperature still rises, so the increase in London forces must outweigh the decrease in dipole–dipole forces. More energy is needed to separate the molecules.
4. **HF** has the fewest electrons, yet it has the highest boiling temperature, because HF molecules form **hydrogen bonds** (H bonded to F, with lone pairs on F). Hydrogen bonds are much stronger than the other intermolecular forces, so far more energy is needed to overcome them.

Solvents and solubility

For a substance to dissolve, its particles must be separated from each other, and the solvent molecules must be pushed apart to make room. Both steps need energy. That energy is supplied by forming new attractions between the solute particles and the solvent molecules. A substance dissolves well when the new solute–solvent attractions are similar in strength to the ones that were broken.

Ionic compounds in water

Water is very polar. When an ionic solid dissolves, water molecules cluster round each ion: the δ− oxygen atoms point towards positive ions, and the δ+ hydrogen atoms point towards negative ions. The ions are said to be **hydrated**. The energy released by these ion–dipole attractions helps to pay for breaking up the ionic lattice. Small ions and ions with larger charges (higher charge density) attract water molecules more strongly and are more strongly hydrated.



δ− oxygen atoms face the positive ion

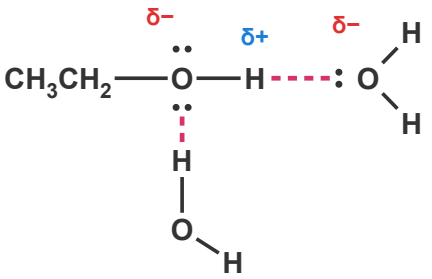
δ+ hydrogen atoms face the negative ion

Hydrated ions. The water molecules turn so that their oppositely charged end faces the ion. The attraction between an ion and these polar molecules releases the energy that breaks up the lattice.

Alcohols in water

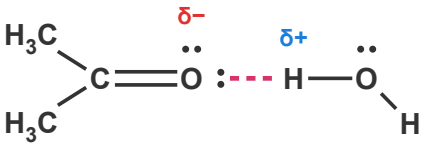
Methanol, ethanol and propanol mix with water in any proportions, because their –OH groups form hydrogen bonds with water molecules, replacing the hydrogen bonds that were broken. As the hydrocarbon chain gets longer, solubility falls. The non-polar chain cannot hydrogen bond, and it gets in the way of the hydrogen bonds between water molecules, so hexan-1-ol is only slightly soluble.

ethanol and water



ethanol both gives and accepts hydrogen bonds

propanone and water



propanone can only accept a hydrogen bond

Hydrogen bonds (pink) between water and organic molecules. Ethanol can form them both ways, through its O–H hydrogen and through its oxygen lone pairs. Propanone has no O–H, but water can still hydrogen bond to its oxygen, which is why propanone mixes with water.

Compounds that do not dissolve in water

A compound that cannot form hydrogen bonds with water is usually insoluble, even if it is polar. Halogenoalkanes such as 1-chlorobutane are polar, but the dipole–dipole and London attractions they could form with water are much weaker than the hydrogen bonds between water molecules that would have to break. Water molecules stay hydrogen-bonded to each other, and the halogenoalkane forms a separate layer. Non-polar substances such as hexane are insoluble in water for the same reason.

Non-aqueous solvents

Substances tend to dissolve in solvents whose intermolecular forces are similar to their own ("like dissolves like"). Iodine, which has only London forces, dissolves readily in hexane, which also has only London forces, but hardly at all in water. Propanone is a very useful solvent because it has both characters: its CH_3 groups form London forces with non-polar substances, and its polar C=O group attracts polar molecules and accepts hydrogen bonds from water, so it mixes with water too.

EXAM TIP

In a solubility answer, name the forces broken in the solute, the forces broken in the solvent, and the new forces formed between them, then compare their strengths. "Hydrogen bonds form between the –OH group of ethanol and water molecules" earns more than "ethanol is polar".

Quick checks

QUICK CHECK

What happens when liquid bromine, Br_2, boils?

- Br–Br covalent bonds break, forming bromine atoms.
- London forces between Br_2 molecules are overcome.
- Hydrogen bonds between Br_2 molecules are broken.
- Permanent dipole–dipole forces between Br_2 molecules are overcome.

Boiling separates whole molecules, so no covalent bonds break. Br_2 is non-polar, so the only intermolecular forces are London forces.

QUICK CHECK

Why do the boiling temperatures of the noble gases increase from helium to xenon?

- The atoms become more polar.
- The covalent bonds between the atoms get stronger.
- The atoms have more electrons, so the instantaneous and induced dipoles are larger and the London forces stronger.
- Heavier atoms move more slowly, so they cannot escape from the liquid.

Noble gas atoms are single, non-polar atoms with no covalent bonds between them. Their only attraction is London forces, which increase with the number of electrons.

QUICK CHECK

Which of these substances forms hydrogen bonds between its own molecules?

- fluoromethane, CH_3F
- propanone, CH_3COCH_3
- methylamine, CH_3NH_2
- hydrogen sulfide, H_2S

Methylamine has H atoms bonded to nitrogen, which also carries a lone pair. In CH_3F and propanone no H is bonded to N, O or F; sulfur is not electronegative enough for H_2S to hydrogen bond.

QUICK CHECK

Which statement about hydrogen bonding in ice is correct?

- Each water molecule is hydrogen-bonded to two others.
- The hydrogen bonds hold the molecules further apart than in liquid water, so ice is less dense.
- The O–H···O angle at each hydrogen bond is 104.5°.
- When ice melts, the covalent O–H bonds break.

In ice each molecule is hydrogen-bonded to four others in an open tetrahedral lattice. The O–H···O arrangement is linear (180°); 104.5° is the H–O–H bond angle inside one molecule. Melting only breaks some hydrogen bonds.

QUICK CHECK

HCl boils at 188 K and HBr at 206 K. Which explanation is correct?

- HBr is more polar, so its dipole–dipole forces are stronger.
- HBr forms hydrogen bonds but HCl does not.
- The H–Br bond is stronger than the H–Cl bond.
- HBr has more electrons, so its London forces are stronger, outweighing its smaller dipole.

Bromine is less electronegative than chlorine, so HBr is actually less polar. Neither forms hydrogen bonds, and bond strength inside the molecule is irrelevant to boiling. The extra electrons strengthen the London forces enough to raise the boiling temperature.

QUICK CHECK

Butane (34 electrons) boils at −1 °C and propan-1-ol (34 electrons) at 97 °C. What is the main reason?

- Propan-1-ol molecules form hydrogen bonds; butane molecules only have London forces.
- Propan-1-ol has more electrons, so stronger London forces.
- Butane is branched, so its molecules have less contact.
- Propan-1-ol has stronger covalent bonds.

The two have the same number of electrons, so their London forces are similar. The difference is the hydrogen bonding through the –OH group, which needs much more energy to overcome. Butane is a straight chain, not branched.

QUICK CHECK

Why is 1-chlorobutane almost insoluble in water, even though its molecules are polar?

- It reacts with water.
- Its molecules are too large to fit between water molecules.
- It cannot form hydrogen bonds with water, and its weaker attractions to water do not make up for breaking hydrogen bonds between water molecules.
- Its C–Cl bond is non-polar.

To dissolve, it would have to break into water's hydrogen-bonded network. The dipole–dipole and London attractions it can form with water are too weak to compensate, so it stays as a separate layer.

QUICK CHECK

In a solution of sodium chloride, how are the water molecules arranged around the ions?

- H atoms point towards Na^+; O atoms point towards Cl^-.
- O atoms point towards Na^+; H atoms point towards Cl^-.
- O atoms point towards both ions.
- Water molecules form covalent bonds to both ions.

The oxygen of water is δ−, so it is attracted to the positive sodium ion; the hydrogens are δ+, so they are attracted to the negative chloride ion. These ion–dipole attractions are what hydration means.